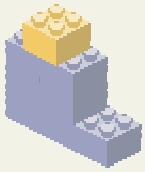
Thomas Scholtz

Technical Product Manager, Aspiring Entrepreneur
Interested in blockchain-based systems in the South African context, especially within the informal sector.

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ABOUT ME

I am a first-principles problem-solver with strengths in product intuition, creativity, ideation and strategic foresight. I have spent my career developing my ability to execute in order to support these strengths. I prefer direct, open communication and am motivated by shared purpose and a meaningful mission. I chose distributed systems as my path into blockchain, which I see as a way to program incentives and encode change. Through my MSc I am specialising in blockchain for South Africa, aiming to build markets that cut information asymmetry, create market efficiency, formalise hidden markets and connect people to opportunity. My experience has taught me that the real world is not all 1s and 0s, and relationships run the world, no matter how formal or informal the setting.

EDUCATION

- **Studying MSc Computer Science**
  How can blockchain-based markets coordinate South Africa's economy?
  University of Stellenbosch
  2026 – Present
- **UZH University of Zurich**
  International Summer School: Deep Dive into Blockchain – Linking Economics, Technology and Law
  Awarded most critical thinker
  5 – 24 July 2026
- **BSc Computer Science Honours, *cum laude***
  University of Stellenbosch
  2021
- **BCom Mathematical Sciences: Data Science**
  University of Stellenbosch
  2018-2020

Courses Blockchain Essentials: An Introduction to Blockchain (ConsenSys), Go: The Complete Developer's Guide, gRPC and Golang Master Class: Modern API and Microservices, Machine Learning A-Z: Python in Data Science, Docker Essentials, Advanced Terraform, Learning Jenkins, Applied Concurrency in Go, Go Design Patterns, Serverless and Microservices for AWS, Foundations of Project Management, Manage Bigtable on Google Cloud

EXPERIENCE

- **Alis Exchange** — Google Cloud Partner
  Engineer, Project & Product Lead • Hybrid, Stellenbosch
  Jun 2023 – Oct 2026 • Protobuf, Go, gRPC, GCP, RBAC, A2A, ADK
  - **Technical Project Lead** *(part-time, Jun – Oct 2026)* – delivered two French-first AI systems for Cactus, a Luxembourg-based retail chain: Market Intelligence (competitor promotions to SKU-level insights) and Procurement Intelligence (SARIMAX sales forecasts to derive explainable daily orders). Unforeseen events shortened timelines to 12 weeks in parallel and a solo delivery which yielded good learnings, including the design of company AI engineering methodology.
  - **Solutions Architect** *(Jun 2025 – Jun 2026)* – in teams of 1–3, developed custom software solutions for clients in wealth management: Portfolio Guidelines Review system for Standard Bank, Financial Data & Reporting for Rezco and Mentenova. Integral to these solutions were patterns which I product managed at various stages: Files Exchange (event-driven file parsing and ingestion), Workflows Exchange (human-in-the-loop AI cron workflows), Reporting Exchange (report pipeline generating various artefacts).
  - **Product Lead** – CodeBlocks, an Alis Build resource for a monetised economy of shared code as distributed Git repos.
  - **Solutions Engineer** *(May 2024 – Jun 2025)*
  - **Platform Engineer** *(Jul 2023 – May 2024)*
- **{^_^} Interface Exchange**
  Founder • collaborative & solo passion projects
  Apr 2025 – Present
  - j2e.app – Interface's first commercial solution, for non-profit work4aliving.org: I lead stakeholders and development of Journey2Excellence, a cross-platform LMS for low-resource environments.
  - testify – Hedera-based verifiable credentials make the job market a public order book where matches between credentialed profiles and open positions earn interviews.
  - episode08.com – social media that is truly social.
- **LifeQ**
  Golang Developer • Hybrid, Stellenbosch
  Sep 2022 – Jun 2023 • Go, Scala, Terraform, Docker, AWS
  In a team of 4, shipped microservices responsible for processing patient data streams from wearables.
- **Octoco**
  Software Engineer • Hybrid, Stellenbosch
  Jan – May 2022 • .NET, Python, Flutter, Firebase, AWS IoT & ML
  Full-stack development of solutions for local start-ups.

PROJECTS

- **iTeksi!: visibility for Cape Town's minibus taxi network**
  As a team of 3 we built iTeksi! at the Builders' Table. Cape Town's 830 000 daily taxi trips are invisible: can we change that to improve South Africa's transport network? Our approach: digitise demand and resolve information asymmetry while minimising change management. Drivers get simple, bold actions that measure what already happens; riders have one proactive signal, a ping showing where they are and where they want to go. Together these feed an insightful dashboard for owners, the persona we sell to: demand by route, vehicle activity and, above all, live earnings. My favourite feature: pay-as-you-go credits redeemed by the driver's QR scan, so vulnerable passengers don't carry cash.
  Built on the Alis Exchange platform
- **Blueprint: A recursive multi-agent system for decomposing concepts into components of a technology stack**
  A deterministic orchestrator recursively invokes three specialised LLM agents — *concept-to-component*, *concept-decomposer*, and *technologist* — which together break a high-level software concept into sub-concepts and map each to a component of a recommended technology stack. The orchestrator itself gates termination via confidence scores on every mapping: either the blueprint is returned or decomposition continues. I am testing the hypothesis that replacing broad task management with a confidence-gated recursive loop over narrow agents lets LLMs handle requests whose scope would otherwise overwhelm a single model.
  Go, A2A, ADK, gRPC, Gemini, Flutter
- **Anomaly Detection in CCTV Surveillance**
  This work proposes a solution to anomaly detection in CCTV surveillance in the form of a consensus framework that combines inputs from three sources, two of which are novel approaches. Experiments were conducted on the UCF Crime data set which comprises 128 hours of untrimmed, public surveillance.
  Python, Tensorflow, Docker, Streamlit `[view paper]`
- **lifeq-cli**
  I accidentally rewrote LifeQ's CLI. While dealing with the consequences of technical debt, I learned lessons about scope creep and preoptimisation.
  Go, Cobra CLI
- **More**
  I've developed a solid understanding of the first principles of computer science via the completion of projects in artificial intelligence, computer vision, advanced algorithms, machine learning, functional programming, networking, parallel computing and software engineering.

MORE ABOUT ME

- **Ndisafunda ukuthetha isiXhosa**
  South Africa has the warmest, most vibrant and expressive cultures. Learning to speak isiXhosa is my way of enjoying that. I often have basic isiXhosa conversations with interesting characters that make my day.
- **Research Publication**
  Stellenbosch University
  My Honours research was published in Volume 85 of the EPiC Series in Computing and the Proceedings of the 43rd Conference of the SAICSIT
  2022
- **Member of Golden Key Society**
  Stellenbosch University
  2021
- **Film Photography**
  I enjoy film photography for a number of reasons: the authenticity of capturing the light of a moment, admiring the thoughtful design of film cameras, delayed gratification, and a reason to travel and explore. `[t0mf00l3ry]`
- **Major Tom & EPISODE^{08}**
  I love a good t-shirt. At university, a friend (Jono Major) and I founded *Major Tom* — a brand built around comic-book artworks I drew in MS Paint and printed on the back of shirts, following a high-profile astronaut and his low-key alien friend through braai scenes laced with global conspiracy theories. EPISODE^{08} was born out of the overlap of my film photography and good t-shirt interests.
- **Outdoor Sports**
  Outdoor endurance sports — trail running, rowing, hiking and surfing — are a great way for me to spend free time. Surfing doesn't come across as an endurance sport, but if you ask me, it is. I also love tennis, although I'm not a natural.
- **SA Schools' Men's 4x Champion**
  SA Schools' National Regatta, Roodeplaat Dam
  2017
- **SA Schools' Boatrace Champion**
  Annual SA Schools' Boatrace, Kowie River
  2015
  `[Item(8): An account by Theo Garrun]`
- **An AI Paradox**
  I'm currently living with a paradox: I am highly skeptical whether AI will have a net positive effect on humanity, but I also spend a significant amount of time working with AI and building AI agents. I definitely think AI is not all good and not all bad — it depends on the motives of its developers and operators.

